VISHNU TEJA KUNDE

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Skills

Frameworks	Python, PyTorch, GitHub, Hugging Face, Diffusers, Wandb, TRL, Accelerate, PEFT, LoRA
Areas of Expertise	Agentic AI, RL, diffusion LLMs, diffusion/flow models

Selected Publications

Vishnu Teja Kunde*, Fatemeh Doudi*, et al. **Reinforcement Learning** for **Diffusion LLMs** with Entropy-Guided Step Selection and Stepwise Advantages. *To appear in* **NeurIPS 2026**. [Paper] [Code]. *Equal contribution.
Tools/Skills: Diffusion LLMs, RL finetuning, policy gradient techniques, variance reduction.

- Formulated reasoning as multi-step MDP over latent states in masked diffusion LLMs.
- Reduced training compute by taking policy gradients only on high-entropy steps, with one-step greedy denoised value estimates for finer credit assignment; state-of-the-art on almost all benchmarks.

Vishnu Teja Kunde*, Mahdi Farahbakhsh*, et al. **Inference-Time** Search using Side Information for **Diffusion**-based Image Reconstruction. *To appear in* **NeurIPS 2026**. [Paper] [Code]. *Equal contribution.
Tools/Skills: PyTorch, diffusion models, deep learning, probabilistic methods, Bayesian inference.

- Developed a principled, training-free, inference-time search algorithm using side information that significantly improves diffusion-based reconstruction for inverse problems.

Mahdi Farahbakhsh, Ilan Sela, Fatemeh Doudi, **Vishnu Teja Kunde**, et al. **AgentDiscover**: Autonomous Discovery with Minimal Search Scaffolding. *arXiv preprint*, 2026. [Paper] [Code].
Tools/Skills: Agentic AI, coding agents, LLMs, long-term memory, search.

- An agent that plans its own search with a database of attempts as long-term memory; beats prior frameworks at lower cost.

Professional Experience

- **Summer Intern**, Qualcomm AI Research, San Diego: developed an agentic harness for multi-hop deep retrieval. May-Aug 2026

Research Projects

- **LLM for compression**: LoRA fine-tuned LLaMA-3.2-1B for text compression, improving rates by 1.5-5%. 2025
- **Transformers & in-context learning**: Trained a transformer to predict HMM belief states, matching Baum-Welch. 2024

Education

- **Ph. D.**, Computer Engineering, ECEN, Texas A&M University, USA 2022–
- **M. Tech.**, Signal Processing, ECE, Indian Institute of Science (IISc), India 2022
- **B. Tech.**, EEE, National Institute of Technology (NITW), Warangal, India 2020

Relevant Coursework

Reinforcement learning, deep learning, high dimensional probability, convex optimization, information theory, measure theory.

Professional Service and Honors

- Gold Reviewer, ICML 2026
- Reviewer, NeurIPS, ICLR, ICML
- Masters fellow, CNI, IISc
- Merit scholarship, NITW, 2016-20
- AIR-179, GATE-EE, 2020
- AIR-6646, JEE-Mains, 2016